Q1 JEE Main 2020 - 2 September (Evening)

When the temperature of a metal wire is increased from $0^{\circ} \mathrm{C}$ to $10^{\circ} \mathrm{C}$, its length increases by 0.02%. The percentage change in its mass density will be closest to

(A) 2.3
(B) 0.06
(C) 0.8
(D) 0.008

Q2 JEE Main 2020 - 3 September (Morning)

A bakelite beaker has volume capacity of $500 \mathrm{cc}$ at $30^{\circ} \mathrm{C}$. When it is partially filled with V_{m} volume (at $30^{\circ} \mathrm{C}$) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(\text{beaker})}=6 \times 10^{-6\circ} \mathrm{C}^{-1}$ and $\gamma_{(\text{mercury })}=1.5 \times 10^{-4\circ} \mathrm{C}^{-1}$, where γ is the coefficient of volume expansion, then V_{m} (in cc) is close to

Q3 JEE Main 2020 - 3 September (Evening)

A calorimeter of water equivalent $20 \mathrm{g}$ contains $180 \mathrm{g}$ of water at $25^{\circ} \mathrm{C}$. 'm' grams of steam at $100^{\circ} \mathrm{C}$ is mixed in it till the temperature of the mixture is $31^{\circ} \mathrm{C}$. The value of $' \mathrm{m} '$ is close to (Latent heat of water $=540$ cal g^{-1}, specific heat of water $=1$ cal $\left.\mathrm{g}^{-1\circ} \mathrm{C}^{-1}\right)$

(A) 2
(B) 3.2
(C) 2.6
(D) 4

Q4 JEE Main 2020 - 4 September (Morning)

The specific heat of water $=4200 \mathrm{Jkg}^{-1} \mathrm{K}^{-1}$ and the latent heat of ice $=3.4 \times 10^{5} \mathrm{Jkg}^{-1}$ 100 grams of ice at $0^{\circ} \mathrm{C}$ is placed in $200 \mathrm{g}$ of water at $25^{\circ} \mathrm{C}$. The amount of ice that will melt as the temperature of water reaches $0^{\circ} \mathrm{C}$ is close to (in grams)

(A) 69.3
(B) 63.8
(C) 64.6

(D) 61.7

Q5 JEE Main 2020 - 5 September (Morning)

A bullet of mass 5 g, travelling with a speed of $210 \mathrm{m/s}$, strikes a fixed wooden target. One half of its kinetic energy is converted into heat in the bullet while the other half is converted into heat in the wood. The rise of temperature of the bullet if the specific heat of its material is $0.030 \mathrm{\,cal} / (\mathrm{g} - {}^\circ\mathrm{C})$ $\left(1 \mathrm{cal} = 4.2 \times 10^{7}\right.$ ergs) close to

(A) $83.3^\circ\mathrm{C}$
(B) $87.5^\circ\mathrm{C}$
(C) $38.4^\circ\mathrm{C}$
(D) $119.2^\circ\mathrm{C}$

Q6 JEE Main 2020 - 5 September (Evening)

Two different wires having lengths L_1 and L_2, and respective temperature coefficient of linear expansion α_1 and α_2, are joined end-to-end. Then the effective temperature coefficient of linear expansion is

(A) $\sqrt[2]{\alpha_1 \alpha_2}$
(B) $4 \frac{\alpha_1 \alpha_2}{\alpha_1+\alpha_2} \frac{\mathrm{L}_2 \mathrm{L}_1}{(\mathrm{L}_2+\mathrm{L}_1)^2}$
(C) $\frac{\alpha_1+\alpha_2}{2}$
(D) $\frac{\alpha_1 L_1+\alpha_2 L_2}{L_1+L_2}$

Q7 JEE Main 2020 - 8 January (Evening)

There are three containers $C_1,\ C_2$ and C_3 filled with same material at different constant temperature. When we mix then for different volume then we get some final temperature as shown in the below table. So find value of final temperature θ as shown in the table.

C_1	C_2	C_3	$t\,(^\circ C)$
1ℓ	2ℓ	0	60
0	1ℓ	2ℓ	30
2ℓ	0	1ℓ	60
1ℓ	1ℓ	1ℓ	θ

Answer Key

Q1 (B)	**Q2** (20)	**Q3** (A)	**Q4** (D)
Q5 (B)	**Q6** (D)	**Q7** (50)	

Q1

$\rho=\frac{\mathrm{M}}{\mathrm{V}} \Rightarrow\left|\frac{\Delta \rho}{\rho} \times 100\right|=\left|\frac{\Delta \mathrm{V}}{\mathrm{V}} \times 100\right.$

$\left|\frac{\Delta \rho}{\rho} \times 100\right|=3 \alpha \Delta \mathrm{T} \times 100$

Given $\frac{\Delta \ell}{\ell}=2 \times 10^{-4} \quad \Rightarrow \alpha \Delta \mathrm{T}=2 \times 10^{-4}$

$\Rightarrow \alpha=2 \times 10^{-5}$

From (i), $\frac{\Delta \rho}{\rho} \times 100=6 \times 10^{-5} \times 10 \times 100$

$=0.06$

Q2

$V_0=500 \mathrm{cc}$

$\mathrm{v}_{\mathrm{b}}=\mathrm{v}_0+\mathrm{V}_0 \gamma_{\text {beaker }} \Delta \mathrm{T}$

And for Mercury

$\mathrm{V}_{\mathrm{B}}^{\prime}=\mathrm{V}_{\mathrm{m}}+\mathrm{V}_{\mathrm{m}} \gamma_{\mathrm{m}} \Delta \mathrm{T}$

Unfilled volume $\left(V_0-V_m\right)=\left(V_b-V_m^{\prime}\right)$

$\Rightarrow \mathrm{V}_0 \gamma_{\text {beaker }}=\mathrm{V}_{\mathrm{m}} \gamma_{\mathrm{M}} \quad \therefore \mathrm{V}_{\mathrm{m}}=\frac{500 \times 6 \times 10^{-6}}{1.5 \times 10^{-4}}$

$\Rightarrow V_{\mathrm{m}}=20 \mathrm{cc}$

Q3

$\mathrm{m}[540+(100-31)]=200 \times[31-25]$

$\mathrm{m}=\frac{1200}{609}$

$\simeq 2 \mathrm{gm}$

Q4

MS $\Delta T=m L$

$\Rightarrow \quad \mathbf{m}=\frac{\mathbf{M S} \Delta \mathbf{T}}{\mathbf{L}}$

$=\frac{(200)(4200)(25)}{3.4 \times 10^5} \mathrm{gm}$

$=61.7 \mathrm{gm}$

Q5

$$\mathrm{MS\Delta T} = \frac{1}{2}\left(\frac{1}{2}\mathrm{Mv}^2\right)$$
$$\Rightarrow \Delta T = \frac{v^2}{4S}$$
$$= \frac{(210,00)^2}{4 \times 0.030 \times 4.2 \times 10^7}$$
$$\simeq 87.5^\circ \mathrm{C}$$

Q6

$$(\mathrm{L}_1 + \mathrm{L}_2)\,\alpha_{\mathrm{eq}} \times \Delta\mathrm{T} = \mathrm{L}_1\alpha_1\Delta\mathrm{T} + \mathrm{L}_2\alpha_2\Delta\mathrm{T}$$
$$\Rightarrow \alpha_{\mathrm{eq}} = \frac{\mathrm{L}_1\alpha_1 + \mathrm{L}_2\alpha_2}{(\mathrm{L}_1 + \mathrm{L}_2)}$$

Q7

$$1\theta_1 + 2\theta_2 = (1+2)\,60$$
$$\theta_1 + 2\theta_2 = 180 \qquad \ldots(1)$$
$$0 \times \theta_1 + 1 \times \theta_2 + 2 \times \theta_3 = (1+2)\,30$$
$$\theta_2 + 2\theta_3 = 90 \qquad \ldots(2)$$
$$2 \times \theta_1 + 0 \times \theta_2 + 1 \times \theta_3 = (2+1)\,60$$
$$2\theta_1 + \theta_3 = 180 \qquad \ldots(3)$$
$$\text{and } \theta_1 + \theta_2 + \theta_3 = (1+1+1)\,\theta \qquad \ldots(4)$$
$$\text{from } (1) + (2) + (3)$$
$$3\theta_1 + 3\theta_2 + 3\theta_3 = 450 \;\Rightarrow\; \theta_1 + \theta_2 + \theta_3 = 150$$
$$\text{from } (4) \text{ equation } 150 = 3\theta \;\Rightarrow\; \theta = 50^\circ C$$